



Automatically Generated Cardiovascular Digital Twin in Critical Care: A Proof of Concept Study

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Abstract. This proof of concept study demonstrates the capabilities of a virtually automatically generated digital twin framework for enhancing hemodynamic monitoring in critical care. By combining a deterministic cardiovascular model with patient-specific data through data assimilation techniques, the digital twin can act as a data denoiser, reconstruct physiological waveforms that are typically unavailable in critical care settings and generate clinically relevant biomarkers. Validation was performed using real data from patients under general anesthesia. The proposed framework efficient calibration and ability to follow the patient's state over time supports the possibility of real-time bedside applications.

Keywords: Digital twin · Model-data interaction · Cardiovascular · Critical care monitoring

1 Introduction

Hemodynamic monitoring is a cornerstone of critical care, guiding the management of the patient's cardiovascular state. Anesthesiologists and intensivists aim to achieve an optimal balance between the maintenance of an adequate hemodynamic state and minimizing the exposure to unnecessary drugs or overfilling, as both extremes can have detrimental consequences. However, classical monitoring data often fall short of providing conclusive insights into the patients' cardiovascular states and their likely response to therapeutic interventions. Moreover, critical care decisions are frequently made under time constraints, where delayed actions can severely impact the patients' outcome [16, 18]. These challenges highlight a pressing need for innovative and personalized approaches to improve hemodynamic monitoring and decision-making in critical care settings [9, 10].

Digital twins have emerged as a promising solution in this regard. A digital twin serves as a computational model of the patient's cardiovascular system, capable of replicating physiological behavior and offering additional insights into cardiovascular function. Benefiting from their predictive capabilities, digital twins have already demonstrated their potential in enhancing cardiovascular medical practices [2, 3, 5, 6, 8, 20].

In the context of hemodynamic monitoring, a previous work has shown the feasibility of augmenting monitoring frameworks with biophysical models of the cardiovascular system to simulate ventricular pressure-volume loops and infer functional cardiovascular parameters [14]. Building on these results, this study aims to advance the integration of digital twins into clinical practice by automatizing their generation, which is a critical step toward real-world applicability.

To achieve this, we use a sequential estimator, merging real-time data from monitoring devices with a cardiovascular model for the estimation of some model parameters. The remaining significant parameters are set by a human operator following specific rules, making our method virtually automatic. We validate this approach using real-world data from patients under general anesthesia, demonstrating its promising capabilities for signal denoising, signal prediction and biomarker generation and paving the way for practical applications of digital twins in critical care.

2 Data Available

For this study, we have access to two datasets. Ethical clearance for the reuse of these datasets in the present study was granted by the ad hoc ethics committees of the clinical data warehouse of Greater Paris University Hospitals (AP-HP) (IRB00011591, CSE-EDS n°21-42).

The first dataset comes from the AnestHemo clinical trial (CE-SRLF 14-34). It contains waveform data for central arterial pressure measured via a catheter located in the aortic arch, blood velocity measured in the descending aorta with an esophageal Doppler probe (Deltex ODM+), and ECG derivations. The velocity time integral (VTI) was measured using transthoracic echocardiography for each patient before the intervention. Combined with abacuses to estimate aortic radius from patient metadata [22], this allows to reconstruct the ascending aorta blood flow from the descending aorta velocity signal. The waveforms in both datasets are sampled at intervals of 8 ms. 45 patients were enrolled in the AnestHemo clinical study. Among these patients, 16 suffered from a hypotension episode and were treated with an administration of noradrenaline. These patients were included in the hypotensive group. The remaining 29 patients were included in the normotensive group.

The second dataset originates from the Looping clinical trial [1]. It includes data similar to that collected in the AnestHemo clinical trial, excluding the ECG derivations, along with additional left ventricular volume measurements obtained using 3D echocardiography. In the Looping dataset, only a single heartbeat is available for each patient. To generate longer sequences, the single heartbeat signal is extended synthetically by periodic replication. Only one patient of the

Looping dataset has been considered in this study, the Looping dataset serving to show a prospective capability of the digital twin. Current efforts are directed towards extending these findings to a larger part of the Looping cohort.

3 Methods

The digital twin that we develop integrates two key components: (1) a deterministic cardiovascular model that embeds prior knowledge and (2) patient-specific data obtained from continuous monitoring. These components are combined using a data assimilation procedure, which updates the model state and parameters based on the data collected over time. The digital twin is then used to produce additional information on the patient's cardiovascular state.

3.1 Cardiovascular Model Description

The cardiovascular model aims to capture the key elements of physiology that are relevant for hemodynamic management in the operating room and intensive care units and to allow the incorporation the available monitoring data.

Hemodynamic management in these settings is conducted based on a simplified representation of the main cardiovascular compartments, such as the heart and arterial system. Monitoring data typically include blood pressure measurements at various points in the vascular system. Therefore, a lumped representation of the cardiovascular system is adapted to our needs.

In our model, the heart is represented by a single-ventricle with a simplified geometry [4], coupled with a two-stage Windkessel circulation (double RC element) through lumped valves, see Fig. 1. The model's inputs are the ventricular activation function, the time varying atrial pressure, and the venous pressure. Outputs are all combinations of cardiovascular state components, notably the ventricular volume and pressure. Model parameters characterize the geometry of the heart, its active and passive mechanical properties, and the resistive and capacitive properties of the circulatory compartments. A detailed description of the model can be found in [15].

The model can be formalized as a non-linear deterministic finite dimensional dynamical system in the form of

$$\begin{cases} \dot{x}(t) = \Phi(x(t), \beta(t)), \\ x(0) = x_0, \end{cases} \quad (1)$$

where x is the state vector, which is of small dimension (typically 10), and β is the parameter vector.

3.2 Patient Specific Calibration Through a Data Assimilation Procedure

Monitoring devices provide data at regular time intervals that constitute observation data on the dynamical system (1). These data are used to correct sequentially the model state and adjust the value of some model parameters as the

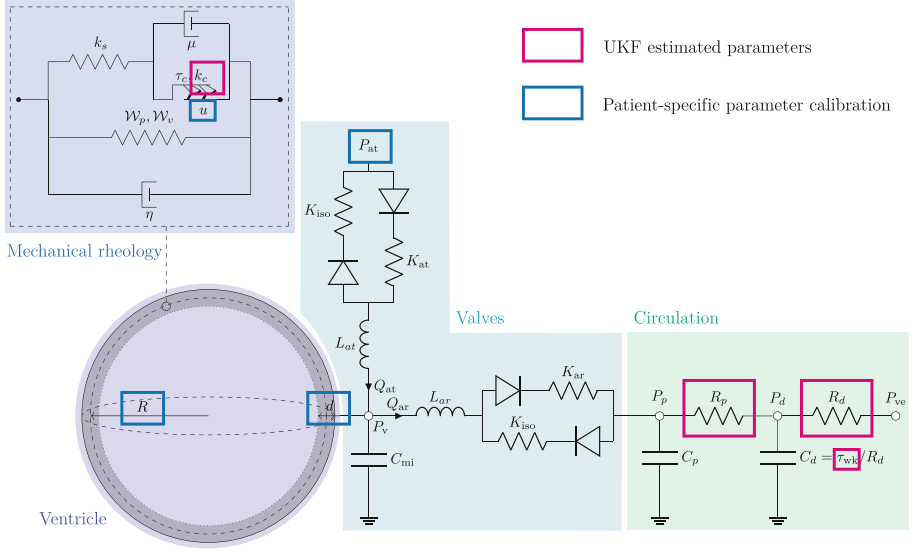


Fig. 1. Model description. We use a lumped representation of the cardiovascular system. The heart is represented by a single-ventricle with a simplified geometry, coupled with a two-stage Windkessel circulation through lumped valves. The parameters estimated with the UKF are boxed in purple; parameters that are patient-specifically calibrated with a specific rule are boxed in blue; other parameters are set to fixed default values. (Color figure online)

data become available through a data assimilation procedure, which builds an estimator of the state and parameters combining model and data [13].

We define an augmented state $z = [x^T, \theta^T]^T$ where θ is a sub-vector of β containing the parameters that are estimated in the data assimilation procedure; we denote by γ the remaining parameter sub-vector. We consider a regular time discretization t_n for $n \in \mathbb{N}$ of time step δt and a numerical scheme $\phi_{\delta t}$ adapted to (1) whose solution at time t_n is denoted by x_n such that $x_{n+1} = \phi_{\delta t}(x_n, \theta_n, \gamma_n)$. A detailed description of an energy-preserving numerical scheme tailored for this model can be found in [15].

The state and parameters are physical quantities that may have intrinsic constraints (e.g. positivity, boundary values). To make sure that the estimation process will not break these constraints and to model the fact that the uncertainty on the quantity may be better represented by an additive Gaussian noise applied to a transformation of the quantity, we apply a bijective transformation g mapping, for each component of the augmented state, from $\mathbb{R}$ to the constrained space [7]. For instance, a positive parameter θ may be associated with its transformed counterpart $\bar{\theta}$ by an exponential function such that $\theta = g(\bar{\theta}) = \exp(\bar{\theta})$. The transformed augmented state discrete dynamics is modelled by the Markov chain

$$\begin{bmatrix} \bar{x}_{n+1} \\ \bar{\theta}_{n+1} \end{bmatrix} = \begin{bmatrix} g^{-1} \circ \phi_{\delta t} \left[g \left(\begin{bmatrix} \bar{x}_n \\ \bar{\theta}_n \end{bmatrix} \right), \gamma_n \right] \\ 0 \end{bmatrix} + \mathcal{W}_n,$$

where $\{\mathcal{W}_n\}_{n \in \mathbb{N}}$ are independent identically distributed Gaussian random variables with $\mathcal{W}_n \sim \mathcal{N}(0, \Sigma_{\text{mod}})$ representing the model disturbance. Note that on the parameter part of the augmented state, this model disturbance accounts for the fact that the parameters value may vary over time as the patient's cardiovascular properties change. We rewrite this system with the initial condition as

$$\begin{cases} \bar{z}_{n+1} = \varphi_{\delta t}(\bar{z}_n, \gamma_n) + \mathcal{W}_n, \forall n \in \mathbb{N}, \\ \bar{z}_0 = \begin{bmatrix} \bar{x}_0 \\ \bar{\theta}_0 \end{bmatrix} + \zeta_0, \end{cases} \quad (2)$$

where $\zeta_0 \sim \mathcal{N}(0, \Sigma_{\text{init}})$ models the uncertainty on the initial condition of the system, in particular on the parameters.

The monitoring data y_n are regularly available at times t_n – possibly through a resampling process. We model the measurement process on the patient with an observation operator h acting on the state and subjected to additive measurement errors $\{\chi_n\}_{n \in \mathbb{N}}$ with $\chi_n \sim \mathcal{N}(0, \Sigma_{\text{obs}})$ such that

$$y_n = h(x_n) + \chi_n. \quad (3)$$

The pair $\{(\bar{z}_n, y_n)\}_{n \in \mathbb{N}}$ is thus a Hidden Markov Model. We aim to compute for all n a sequential estimator $\hat{\bar{z}}_n$ of $\bar{z}_n$ knowing the observations $y_{k, k \in \llbracket 0, n \rrbracket}$ provided by the monitoring devices. In this respect, for computational efficiency, we rely on the Unscented Kalman Filter (UKF) [19]. Note that the model disturbance results in a covariance inflation contribution in the covariance evolution equation at each time step, making the estimation process more robust.

3.3 Generation of Clinically Relevant Biomarkers

The digital twin allows for the reconstruction of non-measured physiological signals, providing new insights for clinicians. However, these signals may not always translate into actionable data for clinical practice. To address this, we also evaluate whether the information generated by the digital twin offers valuable clinical insights into the patient's condition. Given the available data, we assess, as a proxy for that, whether a set of biomarkers derived from the digital twin can show statistical differences between two groups of patients in distinct clinically relevant cardiovascular states. The biomarkers analyzed include patient-specific model parameters, parameters derived from the smoothed pressure-velocity loop: the global afterload angle (GALA), and β -angle [21], and parameters from the predicted pressure-volume loop: the arterial elastance and the ventricular elastance [14].

3.4 Statistical Analysis

We evaluate whether digital twin-derived biomarkers show statistical differences between groups of physiologically different patients in two scenarios: (1) between the normotensive and hypotensive groups, and (2) between the pre- and post-noradrenaline administration states within the hypotensive group.

In the first scenario, the samples are independent. We perform a Mann-Whitney U test for each parameter with the null hypothesis that the parameter distribution in the two groups are identical. Results with p-value ≤ 0.05 are considered significant.

In the second scenario, the samples are dependent since they consist of the same patients in two different hemodynamic states. Here, we use a Wilcoxon signed-rank test for each parameter, with the null hypothesis that the difference between distributions is centered at zero. Results with p-value ≤ 0.05 are considered significant.

4 Results

In this study, we aim to demonstrate the capabilities of our digital twin for augmented hemodynamic monitoring during general anesthesia.

4.1 Creation of the Digital Twin

The first step consists in constructing the digital twin for each patient.

A first set of parameters is determined from the available data using specific rules. The echocardiography data, combined with Klotz's formula [12], are used to determine the reference radius and thickness of the ventricle. The systolic atrial pressure is assigned as a time varying function with a pattern scaled with the patient specific heartbeat duration. Its maximal value is assigned based on a classification of the patient performed by a trained physician using the echocardiography data between two levels of pressure: low (7 mmHg) and high (15 mmHg), following recommendations [17]. The passive material parameters are then determined, through a single parameter variation, ensuring that the ventricle's end-diastolic volume at the determined systolic atrial pressure matches the volume observed in the echocardiography data. The ventricle activation is imposed through an activation function u , which represents the rate of cross-bridge formation – when the function is positive – or cross-bridge destruction – when the function is negative. This function is calibrated using the ECG when this data is available, as shown in Fig. 2. Otherwise, it is adjusted manually based on the arterial pressure profile. An ECG waveform is available for all patients in the AnestHemo cohort, which is the primary cohort used in this work. However, it is not available for patients in the Looping cohort, which is used solely to demonstrate the signal prediction capabilities of the digital twin (results presented in Sect. 4.3). In both cohorts, no patient was suffering from rhythm disorder, and we use a standardized value of QRS duration of 90 ms. The ECG peak detection was performed manually in this work.

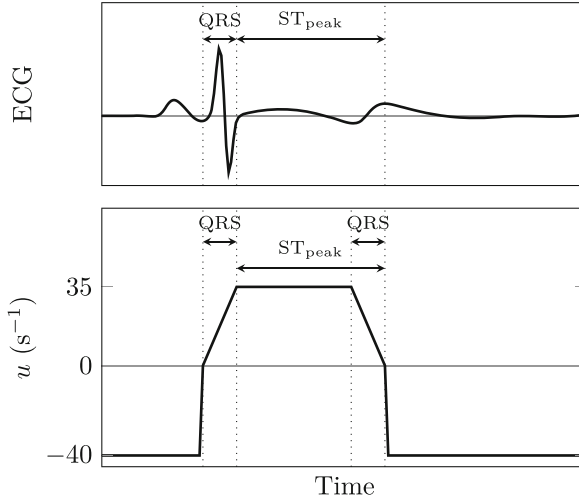


Fig. 2. Illustration of the rule used for the calibration of the activation function u . (Top) Typical ECG signal with the characteristic durations used for the definition of the activation function. (Bottom) Associated activation function. In our study, the QRS duration is standardized, and the activation is solely modulated by the duration of the interval between the end of the QRS-complex and the peak of the T-wave.

A second set of parameters is estimated online from patient-specific data using the data assimilation procedure. Two waveform data types, central arterial pressure and ventricular output blood flow, serve as observations in the UKF for estimating four parameters: the ventricular contractility, the proximal and distal resistances of the two-stage Windkessel model, and the product of the distal resistance and capacitance, which represents the time constant of the Windkessel distal compartment, see Fig. 1.

The UKF estimation is performed on selected stable subsequences of 7 to 9 heartbeats. We set the observation covariance Σ_{obs} so that it corresponds to a relative noise level of $\sim 5\%$ for both signals at typical peak values, the model covariance Σ_{mod} to zero for the state components and so that the standard deviation is about 3% of the typical value for the parameter components and the initial covariance Σ_{init} so that the standard deviation is about 3% of the typical value for the state components and about 100% for the parameter components.

All remaining model parameters, to which the relevant model observables have little sensitivity, are assigned default fixed values.

4.2 The Digital Twin as a Denoising Tool

One key feature of the digital twin is its ability to denoise clinical data by integrating the prior knowledge embedded in the model to the patient data. A typical example is shown in Fig. 3, which illustrates the denoising of the pressure-velocity loop for two patients from the AnestHemo database.

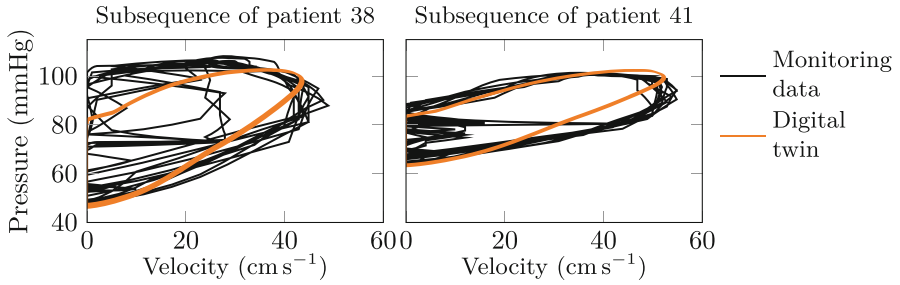


Fig. 3. Illustration of the capability of the digital twin outputs (orange) to denoise the monitoring data (black) for two patients of the AnestHemo dataset. Nine cycles of monitoring data and the corresponding nine cycles of digital twin outputs are presented for both patients. (Color figure online)

4.3 Prediction of Physiological Signals with the Digital Twin

Another capability of the digital twin is its ability to reconstruct physiological waveforms that are not directly available in the data. For illustration, we consider a patient from the Looping trial and her/his corresponding digital twin. While the ventricular volume waveform was measured, it was not used for model calibration. Instead, we compare the measured volume waveform with the predictions generated by the digital twin, see Fig. 4. Note that the original patient monitoring signal contains only a single heartbeat, which has been synthetically replicated to form a multi-beat signal. This may lead to an underestimation of uncertainties in the estimation procedure.

4.4 Statistical Analysis of the Digital Twin Derived Biomarkers

The results of the Mann-Whitney U test performed on the biomarkers in the normotensive and hypotensive groups are presented in Table 1. A statistically significant difference is observed for the ventricular elastance and the distal resistance.

The results of the Wilcoxon signed-rank test performed on the biomarkers in pre- and post-noradrenaline administration states within the hypotensive group are presented in Table 1. A statistically significant difference is observed for all tested parameters except for the Windkessel distal time constant.

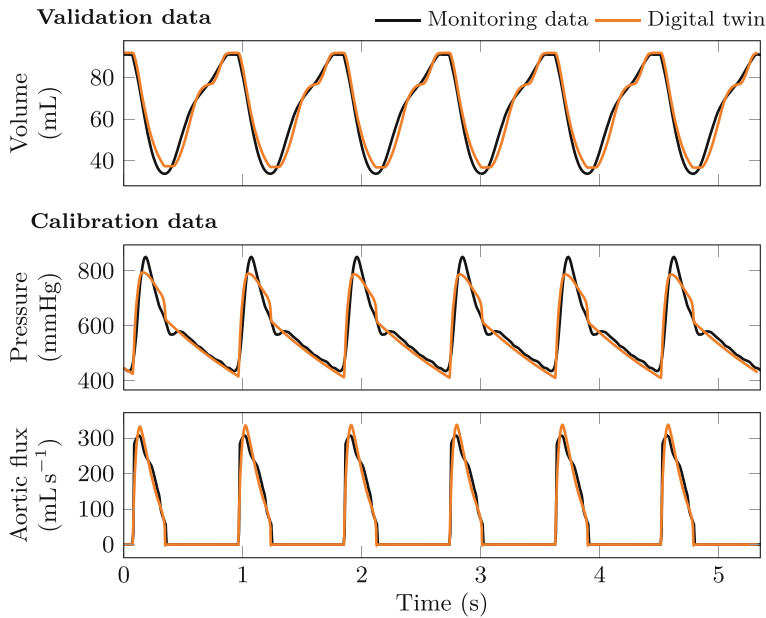


Fig. 4. [Top] Comparison between the volume waveform predicted by the digital twin (orange) and the real measured data (black). [Bottom] Aortic flux and central arterial pressure data used for calibrating the digital twin (orange) and calibrated digital twin outputs (black). (Color figure online)

Table 1. Biomarkers derived from the digital twin for the entire population, and comparison between the normotensive and hypotensive group. Results are expressed as mean $\pm$ standard deviation.

Biomarker	All (n = 45)	Normotensive (n = 29)	Hypotensive (n = 16)	p-value
β -angle ($^{\circ}$)	8.30 ± 8.92	9.83 ± 10.2	5.62 ± 5.04	0.116
Contractility (kPa)	85.8 ± 22.5	88.2 ± 25.2	81.6 ± 16.0	0.457
Elastance arterial (10^8 Pa m^{-3})	1.53 ± 0.686	1.66 ± 0.654	1.29 ± 0.679	0.105
Elastance ventricular (10^8 Pa m^{-3})	4.17 ± 1.93	3.76 ± 1.80	4.90 ± 1.94	0.0495
GALA ($^{\circ}$)	41.1 ± 13.0	42.6 ± 13.9	38.4 ± 10.8	0.400
Resistance distal (10^8 Pa s m^{-3})	1.36 ± 0.403	1.46 ± 0.417	1.20 ± 0.315	0.0308
Resistance proximal (10^8 Pa s m^{-3})	0.164 ± 0.0558	0.157 ± 0.0561	0.176 ± 0.0533	0.247
Time constant windkessel distal (s)	1.73 ± 0.611	1.78 ± 0.697	1.66 ± 0.412	0.951

Table 2. Biomarkers derived from the digital twin for the hypotensive population, and comparison between the pre- and post-noradrenaline administration states. Results are expressed as mean $\pm$ standard deviation.

Biomarker	All (n = 16)	Before noradrenaline administration (n = 16)	After noradrenaline administration (n = 16)	p-value
β -angle ($^{\circ}$)	8.38 ± 7.36	5.62 ± 5.04	11.0 ± 8.26	≤ 0.001
Contractility (kPa)	85.8 ± 16.3	81.6 ± 16.0	90.0 ± 15.5	≤ 0.001
Elastance arterial (10^8 Pa m^{-3})	1.65 ± 1.08	1.29 ± 0.679	2.02 ± 1.36	≤ 0.001
Elastance ventricular (10^8 Pa m^{-3})	4.60 ± 1.84	4.90 ± 1.94	4.31 ± 1.67	0.051
GALA ($^{\circ}$)	44.4 ± 13.2	38.4 ± 10.8	50.5 ± 12.7	0.058
Resistance distal (10^8 Pa s m^{-3})	1.55 ± 0.665	1.20 ± 0.315	1.91 ± 0.729	≤ 0.001
Resistance proximal (10^8 Pa s m^{-3})	0.201 ± 0.0784	0.176 ± 0.0533	0.227 ± 0.0902	0.002
Time constant windkessel distal (s)	1.55 ± 0.426	1.66 ± 0.412	1.44 ± 0.411	0.004

5 Discussion

Our results show that one primary capability of the digital twin is its role as a data denoiser, improving the quality of information extracted from clinical data. For instance, in the case of the pressure-velocity loop (see Fig. 3), the digital twin enables a more reliable estimation of biomarkers that characterize the ventriculo-aortic coupling [11].

Additionally, the digital twin can generate physiological waveforms, such as the ventricular volume (see Fig. 4). The predicted ventricular volume waveform from the digital twin aligns well with the measured data, achieving a relative L2-norm error of 5.67%. This signal is rarely available in routine critical care monitoring. The ability to reliably reconstruct this signal paves the way for generating ventricular pressure-volume loops, which encapsulate valuable insights into the patient’s cardiovascular state. The demonstration of the digital twin’s capability to generate physiological waveforms is limited by the fact that it was performed on a single patient only in this study. Efforts are underway to expand the validation cohort.

Beyond waveform reconstruction, the digital twin also provides clinical insights into patient state through the biomarkers that it generates. As seen in Table 1, the distal resistance is statistically different between the normotensive and hypotensive groups. This result is consistent with the fact that one major cause of hypotension is a deficit of vascular resistance and indicates that vasoplegia was most likely the main source of hypotension in the considered cohort. More surprisingly, the ventricular elastance also appears to be statistically different between the two groups, indicating that some patients may also suffer from a deficit of inotropy. In the comparison between the pre- and post-noradrenaline administration cardiovascular states, all tested parameters except the arterial and ventricular elastances are effectively different between the groups in a statistically significant manner, see Table 2. Noradrenaline is indeed a vaso-

constrictor drug with a mild inotropic effect. It increases vascular resistance and, consequently, the afterload, which is reflected in the GALA, and the arterial stiffness thus increasing the magnitude of wave-reflection, which is associated with the β -angle. The distal Windkessel time constant is the product of the Windkessel distal resistance and capacitance. Noradrenaline is expected to simultaneously increase resistance and decrease capacitance. Therefore, predicting the evolution of this parameter is not straightforward. Our results show a decrease in this parameter, indicating that the effect on capacitance dominates. Arterial elastance is positively coupled with arterial resistances, while ventricular elastance is coupled with contractility. We thus expect both parameters to increase with the administration of noradrenaline. Indeed, we observe such increase, although it is not statistically significant. A larger sample size may be needed to strengthen the results for these parameters. Given that the administration is in bolus form, it is unsurprising to observe significant effects on the cardiovascular system and noticeable differences between the two groups. These effects are effectively captured by the digital twin-derived biomarkers.

While the characterization of hypotension, which is the natural target in this study given the data available, is not the most valuable outcome since hypotension can be directly observed from the pressure signal, it highlights the physiological relevance of these biomarkers and the potential benefits of digital twin-augmented monitoring systems. Future work should validate these capabilities across more diverse hemodynamic states and broader clinical scenarios. In particular, a definitive result regarding the clinical value of the augmented monitoring information provided by the digital twin would need to come from an interventional clinical trial.

The sequential estimator framework enables the digital twin to be updated dynamically as new data become available. This feature allows the digital twin to continuously reflect the patients' evolving state throughout their care journey. Furthermore, the calibration of the digital twin is computationally efficient, taking approximately 10 ms on a standard laptop. In contrast, it would take several hours for a medical expert to perform a similar task manually. This efficiency highlights the potential for real-time implementation at the patient's bedside, offering timely and personalized insights to support clinical decision-making.

Our digital twin generation scheme is in a proof of concept state. It proposes an algorithmic method to construct and use digital twins for cardiovascular monitoring. The core of the model calibration scheme operates in a non-supervised way. However, in our workflow, some stages are still performed by human operator (following well-defined rules). Additional efforts thus need to be conducted to make the generation of the digital twin fully automatic from an implementation point of view.

6 Conclusion

This study proposed a method to generate virtually automatically cardiovascular digital twins for critical care monitoring. The digital twin ability to dynamically

update with new data, reflecting the patient's evolving condition, and its computational efficiency show potential to warrant real-time, bedside implementation for enhanced decision-making. The results highlight key features of our cardiovascular digital twin, including its ability to denoise clinical data, reconstruct physiological waveforms like ventricular volume, which offers insights into the cardiovascular state, and provide clinically relevant biomarkers.

This study demonstrates the general capabilities of our digital twin in developing augmented monitoring tools. Further studies are needed to confirm these results in broader clinical scenarios and with a more hemodynamically diverse patient cohort. To consider clinical applicability, interventional clinical studies will need to be conducted to assess the concrete benefits obtained for patients using this new monitoring approach.

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